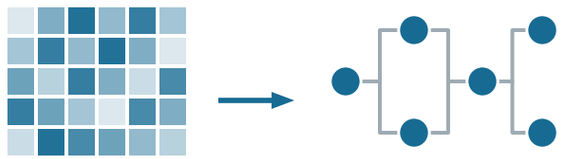
**a** Inputs and frame graph

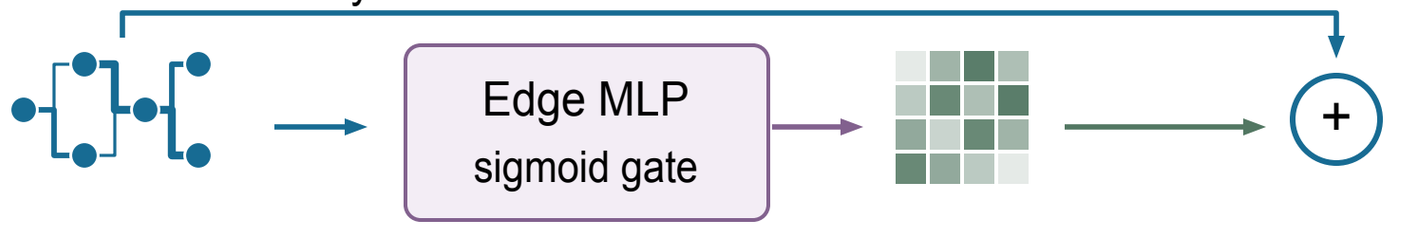


x : q ≤ 50

Within-frame E

No unrelated-field edges

**b** Gated residual layer



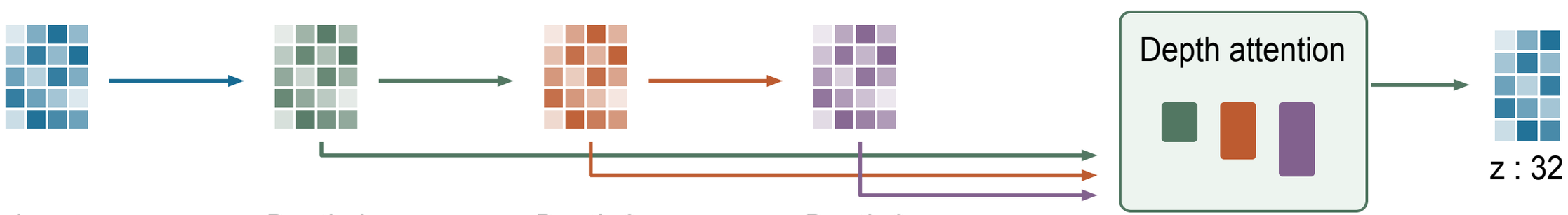

h and distance

Weighted mean

Residual

Normalize by gate sum → ReLU → dropout → LayerNorm

**c** Three depths → attention → embedding




Input

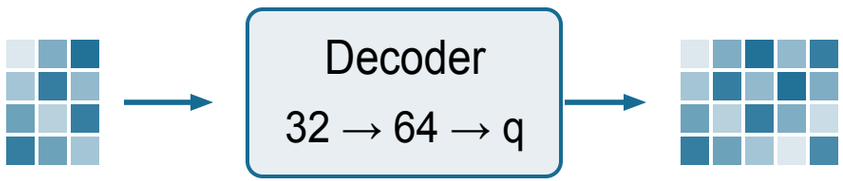
Depth 1

Depth 2

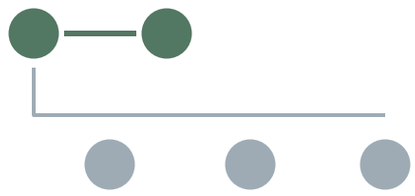
Depth 3

Three 64-dimensional hidden states

**d** Two training objectives




Reconstruction MSE

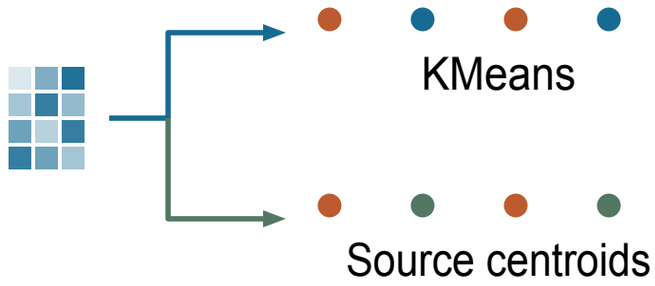


10 random negative draws

Detached gate weights; τ = 0.2

L = reconstruction + 0.5 × contrast

**e** Separate predictions




Seam score ≠ calibrated error